

Real analysis Lecture 7 and 8 homework

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1.1 Let $a, b \in \mathbb{R}$ with $a < b$. Prove $[a, b]$ is closed

The complement of $[a, b]$ is $(-\infty, a) \cup (b, \infty)$, denote the complement by S .

Assume $x < a$, by the density of the Reals, there exists c such that $x < c < a$. Denote $\varepsilon = c - x$, then for every x , $\exists \varepsilon > 0$ such that $x + \varepsilon \in S$. Since $x - \varepsilon < x \Rightarrow x - \varepsilon \in S$.

By the symmetric argument $(x - \varepsilon, x + \varepsilon) \subset S$ if $x > b$, therefore S is open and S^c is closed. $\square$

1.2 Is the set $Z \subset \mathbb{R}$ closed? Provide a proof to substantiate your claim.

Let $x \in Z^c$ and $\varepsilon > 0$ where $\varepsilon = \min(\frac{|x+1|-x}{4}, \frac{x-|x|}{4})$. $\forall a \in (x-\varepsilon, x+\varepsilon)$, $\lfloor x \rfloor < a < \lfloor x+1 \rfloor \Rightarrow (x-\varepsilon, x+\varepsilon) \subset Z^c \Rightarrow Z^c$ is open $\Rightarrow Z$ is closed. $\square$

1.3 Is the set of rationals $Q \subset \mathbb{R}$ closed? Provide a proof to substantiate your claim.

Assume, for contradiction, Q^c is open, it would imply that there exists an $\varepsilon > 0$ such that $(x-\varepsilon, x+\varepsilon) \subset Q^c$ but because of the density of the rationals, there exists $b \notin Q^c$ such that $x < b < x + \varepsilon$. By contradiction, Q is not closed. $\square$

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2.1 a) Let Λ be a set (not necessarily a subset of $\mathbb{R}$), and for each $\lambda \in \Lambda$, let $F_\lambda \subset \mathbb{R}$. Prove that if F_λ is closed for all $\lambda \in \Lambda$ then the set

$$F_\Lambda = \{x \in \mathbb{R} : x \in F_\lambda \forall \lambda \in \Lambda\}$$

is closed.

Since F_λ is closed F_λ^c must be open for all $\lambda \in \Lambda$. Therefore, for any element in $x \in F_\Lambda^c$, $\exists \varepsilon > 0$ st $(x - \varepsilon, x + \varepsilon) \subset F_\Lambda^c$. Since $F_\Lambda^c \subset \bigcup_{\lambda \in \Lambda} F_\lambda^c \Rightarrow (x - \varepsilon, x + \varepsilon) \subset \bigcup_{\lambda \in \Lambda} F_\lambda^c$. Therefore $\bigcup_{\lambda \in \Lambda} F_\lambda^c$ is open, and its complement is closed. By De Morgan's Law $\bigcap_{\lambda \in \Lambda} F_\lambda$ is closed. $\square$

3 Let $F \subset \mathbb{R}$ be a closed set, and let x_n be a sequence of elements of F converging to $x \in \mathbb{R}$. Prove that $x \in F$.

For contradiction, assume $x \in F^c$ since $x_n \rightarrow x$, for $\varepsilon > 0 \exists M \in \mathbb{N}$ st $\forall n \geq M$

$$|x_n - x| < \varepsilon \Rightarrow x - \varepsilon < x_n < x + \varepsilon \Rightarrow x_n \in (x - \varepsilon, x + \varepsilon) \forall \varepsilon > 0$$

Since F^c is open $\exists \varepsilon > 0$ st $(x - \varepsilon, x + \varepsilon) \subset F^c \Rightarrow x_n \in F^c$. Since x_n is both in F and F^c there is a contradiction, and therefore $x \in F$. $\square$

4 Prove that if x_n is a convergent sequence, $k \in \mathbb{N}$, then

$$\lim_{n \rightarrow \infty} x_n^k = \left(\lim_{n \rightarrow \infty} x_n \right)^k$$

This will be proven by induction

Base case: $(k=1) \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_n$

Inductive Step: assume $\lim_{n \rightarrow \infty} x_n^k = \left(\lim_{n \rightarrow \infty} x_n \right)^k$. Since for two convergent sequences a_n and b_n $\lim_{n \rightarrow \infty} a_n b_n = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right)$. Therefore

$$\lim_{n \rightarrow \infty} x_n^{k+1} = \left(\lim_{n \rightarrow \infty} x_n^k \right) \left(\lim_{n \rightarrow \infty} x_n \right) = \left(\lim_{n \rightarrow \infty} x_n \right)^{k+1}$$

By induction $\lim_{n \rightarrow \infty} x_n^k = \left(\lim_{n \rightarrow \infty} x_n \right)^k$ $\square$

5 Let $x_n = \frac{n - \cos(n)}{n}$. Use the squeeze lemma to show that x_n converges and find the limit.

Since $|\cos(n)| \leq 1 \Rightarrow -1 \leq -\cos(n) \leq 1 \Rightarrow \frac{-1}{n} \leq \frac{-\cos(n)}{n} \leq \frac{1}{n} \Rightarrow$

$$1 - \frac{1}{n} \leq \frac{n - \cos(n)}{n} \leq 1 + \frac{1}{n}$$

Since $1 + \frac{1}{n} \rightarrow 1$ and $1 - \frac{1}{n} \rightarrow 1$, by the Squeeze lemma $\frac{n - \cos(n)}{n} \rightarrow 1$

6 . Let $A \subset \mathbb{R}$ be bounded above, and let a_0 be an upper bound for A . Prove that $a_0 = \sup A$ if and only if there exists a sequence a_n of elements of A such that

$$\lim_{n \rightarrow \infty} a_n = a_0.$$

($\Rightarrow$) By assignment 3, if $a_0 = \sup A$ then for all $n \in \mathbb{N}$ there exists $a_n \in A$ such that

$$a_0 - \frac{1}{n} < a_n \leq a_0$$

Since $a_0 - \frac{1}{n} \rightarrow a_0$ and $a_0 \rightarrow a_0$ by the Squeeze theorem $a_n \rightarrow a_0$

($\Leftarrow$) Let $b = \sup A$ we want to show $b = a_0$. Since $b = \sup A$ and a_0 is an upper bound for A $a_0 \geq b$. For contradiction, Assume $a_0 > b$

Since $a_n \rightarrow a_0$ for $\varepsilon > 0 \exists M \in \mathbb{N}$ st $\forall n > M$ Let $\varepsilon = a_0 - b > 0$

$$|a_n - a_0| < a_0 - b \Rightarrow b - a_0 < a_n - a_0 < a_0 - b \Rightarrow b < a_n$$

Since b is the least upper bound, we have a contradiction and $b \leq a_0$, since $b \geq a_0$ and $b = \sup A$, $b = a_0 = \sup A$.

$\square$